

1. Administrative & Study Identification

Full Study Title: A Multicenter, Open-label, Phase 2 Basket Study of MK-5684 in Participants With Selected Solid Tumors (OMAHA-015)
Short Title/Acronym: OMAHA-015 / A Study of MK-5684 in People With Certain Solid Tumors
Protocol Number: MK-5684-015
NCT Number: NCT06979596
Posting ID: 8219245326351789
Company / Sponsor Name: Merck (Merck Sharp & Dohme LLC)
Investigational Product: Opevesostat (MK-5684)
Phase of Development: Phase 2
Trial Status: Recruiting (Currently recruiting participants)
Medical Monitor: MSD Clinical Research Director

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate whether MK-5684 (opevesostat) prolongs Progression-Free Survival (PFS) without the cancer growing or spreading in participants with breast, ovarian, and endometrial cancers compared to standard treatments. * **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Clinical Benefit Rate (CBR) specifically for Cohort A, Overall Survival (OS), and to assess the overall safety and tolerability of MK-5684.

2.2 Background & Rationale MK-5684 (opevesostat) is an oral, non-steroidal, and selective inhibitor of CYP11A1, a key rate-limiting enzyme in steroid hormone synthesis. By blocking the body from making steroid hormones, the study aims to address hormone-driven tumor growth. This basket study evaluates if this novel mechanism can provide superior outcomes for women with specific advanced solid tumors compared to standard-of-care endocrine therapies.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional (Basket Study)
* **Control Type:** Active Comparator (Physician's choice of standard therapy) / Observation (for the ovarian cancer cohort) *
Blinding: Open-label * **Randomization:** Randomized across assigned cohorts

3.2 Planned Enrollment & Duration * **Target \$\$\$:** 250 participants
* **Number of Sites:** 48 global locations * **Estimated Study Start:** 11-Aug-2025 * **Estimated Primary Completion:** 04-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with histologically confirmed diagnosis of primary advanced, recurrent, or metastatic breast, ovarian, or endometrial cancer. 2. For specific cohorts (e.g., Endometrial), treatment naïve or received up to 1 prior line of platinum-based therapy. 3. Human immunodeficiency virus (HIV), Hepatitis B (HBV), and Hepatitis C (HCV) infections are permitted only if well-controlled with undetectable viral loads on appropriate antiviral therapy. 4. Resolved Adverse Events (AEs) from previous anticancer therapies to $\leq$ Grade 1 or baseline.

4.2 Exclusion Criteria 1. Breast cancer that is amenable to treatment with curative intent. 2. Advanced/metastatic symptomatic visceral spread at risk of rapidly evolving into life-threatening complications (e.g., carcinomatous meningitis, significant symptomatic liver metastases, lymphangitic lung metastases). 3. Radiographic evidence of intratumoral cavitation or invasion/infiltration of a major blood vessel.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: MK-5684 administered at 5 mg twice daily. This is co-administered with fludrocortisone acetate (starting at 0.1 mg orally) and dexamethasone acetate (starting at 1 mg orally), both adjusted individually during the study. *** Referenced/Related Steroid Drug Profile:** Cortisone (Trade Names: Cortone, Cortone acetate; ATC Codes: H02AB10, S01BA03). *** Route of Administration:** Oral (Tablet) *** Duration of Treatment:** Continuous until documented disease progression, unacceptable toxicity, or other conditions for study discontinuation are met.

5.2 Concomitant & Prohibited Therapy * Permitted (Control Arms): Physician's choice of standard endocrine therapies based on the cohort. Breast cancer control includes fulvestrant (500 mg) or exemestane (25 mg QD). Endometrial cancer control includes megestrol acetate, medroxyprogesterone acetate, tamoxifen, or letrozole. *** Prohibited:** Any unapproved investigational anticancer agents or treatments not aligned with the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Day -28 to 0	Screening, Informed Consent, Medical History, Baseline Tumor Imaging, Viral Load Labs
2	Day 1 (Cycle 1)	Randomization, First Dose of MK-5684/Control, Baseline Safety check
3	Cycle-based	Interim, Regular treatment cycles (e.g., 28 days), Safety Labs, RECIST 1.1 Tumor Imaging
4	Follow-up	End of Study, Treatment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Progression-Free Survival (PFS) for all cohorts, defined as the time from randomization to the first documented progressive disease (PD) or death due to any cause. PD requires a $\geq 20\%$ and absolute ≥ 5 mm increase in the sum of diameters of target lesions. * **Measurement Method:** Assessed by Blinded Independent Central Review (BICR) using Response Evaluation Criteria in Solid Tumors (RECIST) Version 1.1.

7.2 Secondary & Exploratory Endpoints * **Clinical Benefit Rate (CBR):** Evaluated specifically for Cohort A. * **Objective Response Rate (ORR):** Percentage of participants across all cohorts achieving a confirmed Complete Response (CR) or Partial Response (PR). * **Overall Survival (OS):** Evaluated across all cohorts. * **Duration of Response (DOR):** Time from the first documented CR or PR until disease progression or death. * Evaluation of overall safety, AEs, and tolerability.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous recording of any untoward medical occurrences temporally associated with the study intervention, graded by severity.
 - **Serious Adverse Events (SAEs):** Reported promptly to the sponsor within specified regulatory timelines (typically 24 hours).
 - **Data Monitoring Committee (DMC):** Utilized to monitor patient safety continuously across all basket cohorts.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for all primary efficacy measures (PFS). Safety Set for AE evaluations.
 - **Sample Size Justification:** Target enrollment of $N=250$ to provide adequate statistical power to detect a significant PFS benefit of MK-5684 over standard treatments across the designated tumor cohorts.
 - **Handling of Missing Data:** Follows standard oncology statistical guidelines, utilizing established censoring rules for PFS and OS endpoints.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine clinical site monitoring and data verification by MSD and applicable Contract Research Organizations (CROs).
 - **Audit Trail:** Verified through electronic Case Report Forms (eCRF), Blinded Independent Central Review (BICR) imaging trails, and protocol adherence tracking.
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11. Study Identifiers & Governance

- **Primary ID:** MK-5684-015
 - **Registry Number:** NCT06979596
 - **Posting ID:** 8219245326351789
 - **Trial Status:** Recruiting
 - **Sponsor/Lead Organization:** Merck / Merck Sharp & Dohme LLC
 - **Study Title:** A Study of MK-5684 in People With Certain Solid Tumors (MK-5684-015/OMAHA-015)
 - **Official Regulatory Title:** A Multicenter, Open-label, Phase 2 Basket Study of MK-5684 in Participants With Selected Solid Tumors (OMAHA-015)
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12. Clinical Framework (Oncology/Solid Tumor Specific)

- **Disease Area / Name:** Malignant Neoplasm(s)
 - **Preferred UMLS Name:** Malignant Solid Neoplasm
 - **Condition Definition:** A neoplasm composed of atypical neoplastic, often pleomorphic cells that invade other tissues. Malignant neoplasms often metastasize to distant anatomic sites and may recur after excision. The most common malignant neoplasms are carcinomas, Hodgkin and non-Hodgkin lymphomas, leukemias, melanomas, and sarcomas.
 - **MeSH Code:** D009369
 - **SNOMED ID:** 1240414004
 - **Medical Methodology:** CYP11A1-targeted Steroid Hormone Synthesis Inhibition
 - **Optimization Target:** Prolongation of Progression-Free Survival (PFS) and Objective Response Rate (ORR)
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13. Study Procedures & Methodology

- **Management Pathway:** Standard oncology clinical pathway for advanced/metastatic selected solid tumors
- **Education Protocol (HCP):** Site Initiation Visits, protocol training, and RECIST 1.1 evaluation alignment

- **Education Protocol (Patient):** Informed consent, detailed instruction on oral dosing (MK-5684, fludrocortisone, dexamethasone, or standard-of-care equivalent), and AE diary management
- **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for impartial imaging assessment

14. Observation & Follow-Up Schedule

- **Total Duration:** Estimated 27 months (until Nov 2027) or until disease progression/unacceptable toxicity
- **Onsite Visit Cadence:** Evaluated in standard treatment cycles (e.g., 28-day intervals)
- **Remote Monitoring Frequency:** Routine safety follow-ups per protocol
- **Checklist Review Interval:** Regular tumor imaging assessments for BICR review to track RECIST 1.1 progression

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				